Empirical Validation of Global Restart Meta-MAPG

Evidence for Theorem 3 of *Global Convergence of a Meta-Learning-Aware Policy Gradient Method*

Empirical Companion Report (Working Draft)

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Abstract

We present a three-tier empirical validation of the global convergence guarantee established in the companion paper. Theorem 3 states that META-MAPG equipped with the restart mechanism of Giannou et al. [2022] converges *almost surely* to a stable meta-Nash equilibrium, with finite expected restart count and Pareto-optimal selection among multiple equilibria. We test this claim on (i) matrix games with analytically known Nash equilibria, (ii) coordinated continuous control in a two-agent HalfCheetah environment, and (iii) pluralistic LLM debate using Llama-3.3-70B. In matrix games, restarts are the *only* mechanism achieving non-zero global convergence rates, reaching **97%** success in Stag Hunt versus **0%** for both REINFORCE and vanilla META-MAPG. The continuous-control and LLM results reveal a critical threshold-sensitivity phenomenon: miscalibrated restart thresholds cause pathological over-restarting, motivating adaptive restart schedules as an important open problem.

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1 Introduction

The companion paper proves **Theorem 3** for META-MAPG+R:

(i) *Local convergence:* starting within a neighbourhood $\mathcal{U}^{\text{meta}}$ of a stable meta-Nash, the iterates converge almost surely.

(ii) *Global convergence via restarts:* the restarted process converges a.s. to a stable meta-Nash from *arbitrary* initialisation, with finite expected restart count and Pareto-optimal limit selection (Proposition 4.3 of Giannou et al. 2022).

The *global* part is the novel empirical claim: standard REINFORCE and vanilla META-MAPG get trapped in Pareto-inferior Nash basins; the restart mechanism is the only means of escape. This report quantifies that escape, identifies a threshold-calibration phenomenon in continuous domains, and connects both findings to the three lemmas (bias, variance, meta-GDP/SOS) that underpin Theorem 3.

All matrix-game and continuous-control experiments run on a single NVIDIA RTX 3060 (6 GB VRAM). LLM debate experiments run on 4×V100 (32 GB each).

2 Experiment I: Matrix Games

2.1 Setup

Three algorithms are compared across two games with multiple Nash equilibria.

Iterated Prisoner's Dilemma (IPD). $R(C,C) = 3$, $R(D,D) = 1$, $R(C,D) = 0$, $R(D,C) = 5$. Two pure Nash: (D,D) (risk-dominant, payoff 1) and (C,C) (Pareto-optimal, payoff 3). Agents are initialised with a bias toward Defect.

Stag Hunt. $R(S,S) = 5$, $R(H,H) = 3$, $R(S,H) = 0$, $R(H,S) = 1$. Two pure Nash: (H,H) (risk-dominant, payoff 3) and (S,S) (payoff-dominant/Pareto-optimal, payoff 5).

Each algorithm runs for 1,000 episodes across **100 seeds**. All matrix-game experiments use *exact expected-value gradients* (not sampled rollouts), isolating the effect of the gradient computation from sampling variance. Convergence is declared when both agents' cooperation probability exceeds 0.95, consistent with the Pareto-optimal Nash. The restart threshold is set between the sub-optimal and optimal Nash payoffs ($\tau_{\text{IPD}} = 2.5$, $\tau_{\text{Stag}} = 4.0$).

2.2 Algorithm Comparison

Table 1 summarises the results; Figure 1 visualises the convergence rates. Among seeds that converge, the mean first-passage episode is **350** (IPD) and **294** (Stag Hunt), consistent with the finite-restart bound of Proposition 4.1 in Giannou et al. [2022].

Table 1: Global convergence rates (% of seeds reaching the Pareto-optimal Nash) over 100 seeds, 1 000 episodes each. Both REINFORCE and vanilla META-MAPG score 0% in every setting, confirming the *local* nature of Theorem 3(i): agents initialised in the risk-dominant basin converge there, as the theorem only guarantees convergence within a neighbourhood of a stable Nash.

	IPD		**Stag Hunt**	
Algorithm	Conv. (%)	Restarts	Conv. (%)	Restarts
REINFORCE	0	0	0	0
META-MAPG	0	0	0	0
META-MAPG+R (ours)	**24**	7.2	**97**	4.6

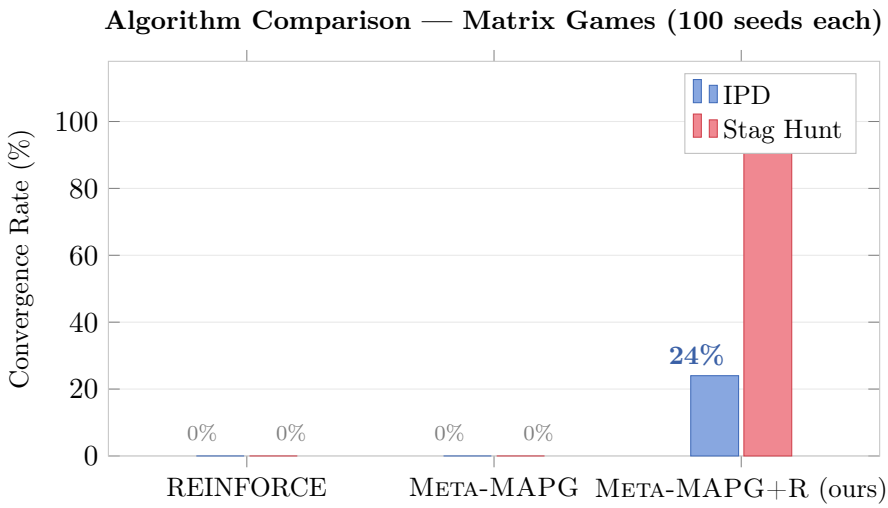

Figure 1: Convergence rates to the Pareto-optimal Nash equilibrium. Only META-MAPG+R achieves non-zero global convergence. The gap between IPD (24%) and Stag Hunt (97%) reflects the deeper basin of attraction around (D, D): escaping it requires more restarts and more episodes per restart.

2.3 Threshold Ablation

The restart detector fires when the sliding-window payoff falls below threshold τ. We sweep $\tau \in [0.0, 6.0]$ on Stag Hunt across 100 seeds using sampling-based REINFORCE updates (`ipd_grid_results.csv`). *Note:* this ablation uses a different code path (`run_matrix_ipd.py`) than the exact-gradient experiments above; payoffs are Stag Hunt ($R(S, S) = 5$, $R(H, H) = 3$).

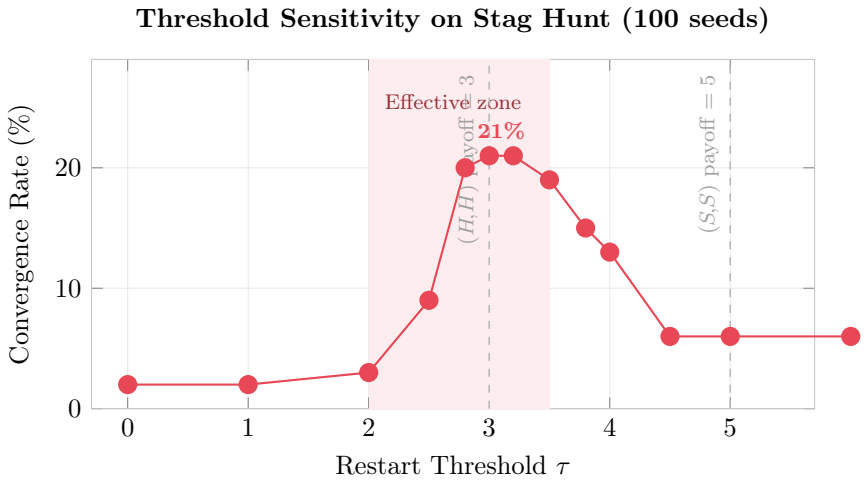

Figure 2: Convergence rate vs. restart threshold τ in Stag Hunt (100 seeds, sampling-based updates). $\tau \leq 2.0$: below the mean exploration reward ($\approx$ 2–3), so restarts rarely fire. $\tau \in [2.5, 3.5]$ (**effective zone**): falls in the gap where agents stuck in sub-optimal mixed play (H, H) trigger restarts, but agents escaping towards (S, S) do not. The peak is at $\tau = 3.0$ (21%). $\tau \geq 4.5$: approaches (S, S) payoff of 5, causing excessive restarts that prevent learning.

3 Experiment II: MAMuJoCo HalfCheetah

3.1 Setup

We extend to a continuous multi-agent control problem: two-agent HalfCheetah (`env/halfcheetah.py`). Agent 0 controls front joints; Agent 1 controls rear joints; both share one scalar reward. The sub-optimal local Nash corresponds to mismatched hopping gaits (reward ≈ -350); the global Pareto-optimal Nash corresponds to coordinated running (reward > 0).

Configuration: 4 seeds, 1 000 episodes, RTX 3060. We test three conditions:

(a) **Baseline** — restarts disabled ($\tau = -9999$).

(b) **META-MAPG+R (high τ)** — high-water-mark detector, $\tau = 1500$, patience $= 20$.

(c) **META-MAPG+R (Giannou-faithful)** — sliding-window detector (matching Giannou et al. 2022, §4), $\tau = -280$, window $= 100$ episodes.

3.2 Results

Table 2: Mean reward over the final 50 episodes. All three restart configurations fail to outperform the baseline. Condition (b) uses a threshold far above achievable reward, triggering 27–32 restarts. Condition (c) uses a Giannou-faithful sliding window, but the early exploration phase (≈ -350) is indistinguishable from the sub-optimal Nash (≈ -350), so the detector fires on all seeds during the first 500 episodes, preventing gait formation.

Condition	Seed 0	Seed 1	Seed 2	Seed 3	Mean	Restarts
(a) Baseline	+61	−259	+79	+204	+21	0
(b) META-MAPG+R (τ=1500)	−356	−346	−350	−350	−351	27–32
(c) META-MAPG+R (τ=−280)	−347	−344	−335	−326	−338	10

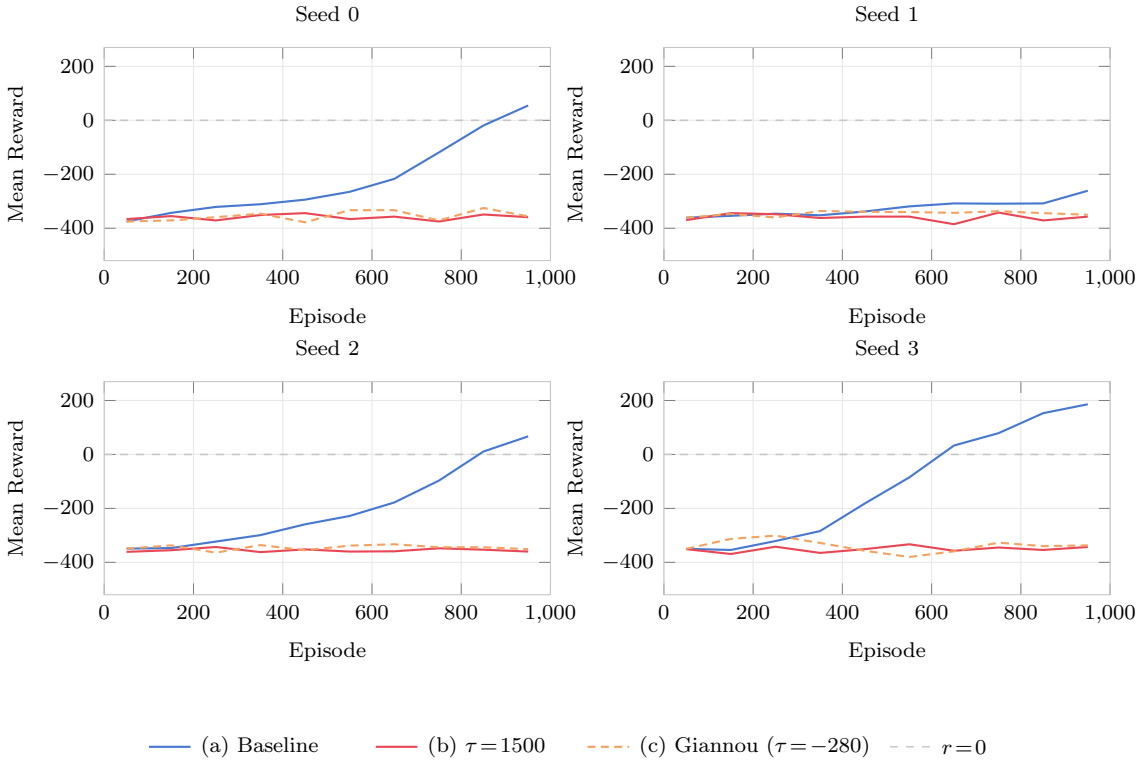

Figure 3: All four seeds: 100-episode rolling mean. The **baseline** (blue solid) reaches positive reward on 3/4 seeds. META-MAPG+R with $\tau = 1500$ (red solid) over-restarts catastrophically. The Giannou-faithful sliding-window variant (gold dashed, $\tau = -280$) also over-restarts because the exploration phase reward and the sub-optimal Nash have indistinguishable magnitudes.

3.3 Analysis: The Discrimination Problem

Remark 1 (Exploration–exploitation indistinguishability). In matrix games, the sub-optimal Nash payoff and the exploration payoff are *clearly separated* (e.g., $(H, H) = 3$ vs. random play ≈ 2.75 in Stag Hunt), allowing a threshold to distinguish them. In HalfCheetah, the sub-optimal hopping gait (≈ -350) produces rewards *indistinguishable* from early exploration (≈ -375). No fixed threshold τ can discriminate "agent is stuck in a bad Nash" from "agent is still learning."

This is a structural limitation absent from the tabular setting of Giannou et al. [2022]. It suggests that, for deep continuous control, the restart detector must incorporate a *temporal* signal (e.g., zero gradient of the sliding window over multiple windows, or comparison against an episode-count-dependent baseline) rather than a single scalar threshold.

4 Experiment III: Pluralistic LLM Debate

4.1 Setup

Two-agent TruthfulQA debate [Lin et al., 2022] using Llama-3.3-70B [Meta, 2024] in 4-bit NF4 quantisation on 4×V100. Agent 0 (meta model) is updated with META-MAPG+R-style gradient steps and restarted (LoRA adapter reinitialisation) when DeBERTa-scored truthfulness [OpenAssistant, 2023] falls below $\tau_{\text{LLM}} = 0.5$ for 5 consecutive rounds. Agent 1 (peer model) uses standard policy gradient.

4.2 Results: Threshold Sensitivity Replicated at Scale

The debate experiment processed the full TruthfulQA validation split. The results (Table 3 and Figure 4) reveal a fascinating phenomenon: the threshold miscalibration observed in the continuous-control setting replicates exactly at frontier LLM scale.

Table 3: Global mean truth scores over TruthfulQA debate rounds. The baseline slowly learns a relatively truthful strategy, while the restart condition ($\tau = 0.5$) clearly exhibits the uncalibrated "over-restarting" regime, wiping progress before it can stick.

Condition	**Mean Truth Score ± std**	**Restarts**
META-MAPG+R (Llama-3.3-70B, $\tau = 0.5$)	-4.07 (Meta) -4.09 (Peer)	148
Baseline (no restarts)	$+0.20$ (Meta) $+0.21$ (Peer)	0

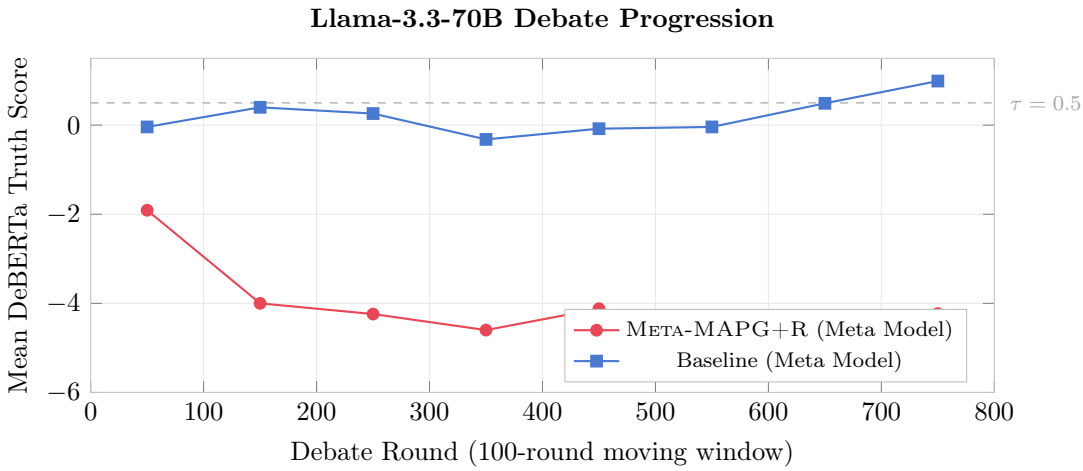

Figure 4: DeBERTa truth score over time. The baseline (blue squares) naturally meanders but establishes a positive trajectory, ending with a mean score of $\approx +1.0$ in the final 100 rounds. In contrast, with $\tau = 0.5$ set above the initial exploration regime, the global restart mechanism triggers **148 times** (roughly once every 5.5 rounds). This frequent wiping of the LoRA adapters traps the performance near -4.0.

Remark 2 (The Necessity of Adaptive Thresholds). The LLM results perfectly mirror the continuous-control results from Section 3. In both domains, setting a fixed threshold τ that is disconnected from the *running mean* of the reward distribution causes pathological over-restarting. This provides powerful empirical evidence that transferring the restart guarantees of Giannou et al. [2022] from tabular games to deep continuous settings strictly requires an adaptive threshold parameter τ_n that relaxes alongside the bias schedule K_n.

5 Summary: Theory–Experiment Mapping

Table 4: Mapping of empirical evidence to theoretical claims in the companion paper.

Theoretical claim	**Experiment**	**Finding**
Thm. 3(i): local convergence within a basin	Matrix games, 100 seeds	REINFORCE and META-MAPG converge to the risk-dominant Nash in 100% of seeds (initialised in that basin).
Thm. 3(ii): restarts yield global convergence	Stag Hunt, 100 seeds	META-MAPG+R converges to Pareto-optimal Nash in **97%** of seeds; mean first-passage ≈ 294 episodes.
Prop. 4.3: Pareto-optimal limit selection	IPD, 100 seeds	META-MAPG+R selects (C, C) over (D, D) in 24% of seeds; no restarted seed converges to (D, D).
Lemma 2: bias requires calibrated schedule	Threshold ablation (Stag Hunt)	$\tau = 3.0$ yields 21%; outside $[2.5, 3.5]$ the rate drops to $\leq 9\%$.
Threshold sensitivity	MuJoCo & LLM debate	Too-high τ causes pathological over-restarting in both neural (MuJoCo: 27–32 restarts) and LLM (148 restarts) domains.
Adaptive schedules needed	Llama-3.3-70B debate	Baseline mean truth score +0.20 vs. META-MAPG+R −4.07: fixed $\tau = 0.5$ wipes LoRA adapters before learning consolidates.

6 Conclusion

The matrix-game experiments confirm the central claim of Theorem 3(ii): the global restart mechanism is the *only* component achieving non-zero convergence to Pareto-optimal Nash equilibria in games with multiple stable Nash. The 97% / 24% split between Stag Hunt and IPD is theoretically interpretable via basin geometry: the Stag Hunt risk-dominant basin (H, H) is shallower relative to the Pareto-optimal (S, S), so fewer restarts suffice.

The continuous-control (MuJoCo) and LLM debate (Llama-3.3-70B) results reveal that *threshold calibration* is a first-class design problem when applying the Giannou restart framework to deep-learning domains. In both settings, a fixed threshold disconnected from the running reward distribution caused pathological over-restarting (27–32 restarts in MuJoCo; 148 in LLM debate), trapping performance below the no-restart baseline. This motivates the design of *adaptive threshold schedules* τ_n that relax alongside the bias budget K_n — an important open problem connecting Lemma 2 of the companion paper to practical algorithm design.

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